

AML Transaction Monitoring Rule-Build Documentation

1. Purpose & Scope

This document outlines the design, data-driven rationale, and validation results for a transaction monitoring rule built on the IBM AML Synthetic Transaction Dataset (5,078,345 transactions, Sep 1–18, 2022). The objective was to identify high-risk transaction patterns indicative of money laundering, specifically layering and mule-account typologies, and to propose a rule structure suitable for analyst review queues.

2. Dataset

Account dataset -

https://www.kaggle.com/datasets/ealtman2019/ibm-transactions-for-anti-money-laundering-aml?select=HI-Small_accounts.csv

Transactions dataset -

https://www.kaggle.com/datasets/ealtman2019/ibm-transactions-for-anti-money-laundering-aml?select=HI-Small_Trans.csv

3. Data Profile Summary

- **Base rate:** 0.10% of transactions are labeled laundering (5,176 of 5,078,345) — consistent with real-world AML base rates, where laundering is a rare-event detection problem.
- **Time window:** 17 days — insufficient for long-window (30-day) behavioral baselines; this constrained Rule design to short-window features.
- **Amount distribution:** Highly right-skewed (median \$1,415; max ~\$1 trillion, likely data artifact). Percentile-based thresholds were used instead of fixed dollar amounts
- **Account Activity:** Hub accounts are clean infrastructure (median 2txns/account; max 168672). Low activity accounts carry more risk

4. Hypotheses Tested and Findings

Hypothesis	Result	Action
Currency mismatch indicates laundering	Disproven — 0.0000% laundering rate at mismatch vs 0.0010% at match	Retired (Rule D)

Cross-bank transfers carry higher risk	Confirmed — 7.8x lift	Retained as flag F1
ACH payment format carries elevated risk	Confirmed — ~7x lift vs base, highest among payment formats	Retained as flag F2
Mid-range amounts (\$3K–\$22K) show elevated risk	Confirmed — risk peaks mid-distribution, declines at extremes	Retained as flag F3
Rare account-pairs (≤3 historical transactions) carry risk	Confirmed — 25x lift, strongest single signal	Retained as flag F4
Low-activity accounts carry risk	Confirmed — 2.8x lift	Retained as flag F5
Time-of-day affects risk	Weak — only ~2x lift	Documented as future enhancement, excluded from MVP to avoid overfitting on a weak signal
Entity fan-out (accounts/banks per entity) is a risk indicator	Disproven — top fan-out entities (e.g. 7,820 accounts/1,185 banks) are financial infrastructure, not laundering networks	Retired
Same-entity (self) transfers indicate layering	Disproven and inverted — 22x lower laundering rate than cross-entity transfers	Retired
Entity-level average risk-score aggregation adds detection coverage	Tested — found entities with 45–92% laundering rates, but contributed 0 incremental recall beyond Rules B/C; fully redundant	Retained as Rule E

5. Rule Design

Rule A — Weighted Composite Score

A linear weighted sum of five behavioral/structural flags, weighted approximately proportional to observed individual lift:

$\text{risk_score} = 5 \times \text{rare_pair} + 3 \times \text{cross_bank} + 3 \times \text{is_ach} + 1 \times \text{mid_amount} + 1 \times \text{low_activity_account}$

- Tier 1 (Investigate): score ≥ 9
- Tier 2 (Priority): score ≥ 11

Rule B — High-Confidence Override

`rare_pair AND cross_bank` → `automatic Tier 2 (Priority)`, applied independent of score, based on evidence of a synergistic (not merely additive) interaction between these two flags (0.733% laundering rate, ~60x base — exceeding the sum of each flag's individual effect).

6. Validation Results

Tier	Volume Flagged	True Positives Caught	Precision	Recall
Tier 1 (Investigate)	0.14%	55	0.80%	1.1%
Tier 2 / Rule B Override	11.4%	4,223	0.73%	81.6%
Combined	11.5%	4,278	0.73%	82.7%

Combined precision represents a **~7.3x lift** over the base laundering rate (0.10%), while capturing 82.7% of all known laundering activity at ~11.5% of total transaction volume.

7. Key Finding — Optimization Opportunity

Rule B (the two-flag override) independently accounts for **81.6%** of all laundering caught, while Rule A's weighted score contributes only **~1.1%** incrementally beyond what Rule B already captures. This indicates the five-flag weighted scoring system adds substantial design and maintenance complexity for marginal recall gain.

8. Limitations Identified

- **Non-monotonic scoring:** Linear additive weighting produced inconsistent risk ordering at intermediate scores (e.g., score=5 showed lower observed risk than score=4), since different flag combinations can collapse to the same total score. A combination/decision-tree-based approach would avoid this collision effect.
- **Short observation window** limited the ability to build longer-horizon behavioral baselines (e.g., 30-day account activity norms).
- **Dataset-specific anomalies:** Wire and Reinvestment payment formats showed 0% laundering rate in this synthetic dataset — inconsistent with typical real-world AML risk priors for wire transfers. This should not be generalized to production rule design without revalidation against real transaction data.

9. Recommendations

Analysis of Rule A's incremental contribution revealed that nearly all of its unique value (beyond Rule B) was concentrated in transactions sharing `rare_pair`, `is_ach`, and `low_activity_account` flags — a same-bank ACH variant of the mule-account typology already captured by Rule B for cross-bank cases. Based on this finding, we recommend retiring the five-flag weighted composite scoring approach (Rule A) in favor of two simple, interpretable combination rules (Rule B and the newly identified Rule C). This reduces system complexity and improves auditability while preserving effectively all detection coverage — a textbook example of rule optimization driven by empirical validation rather than assumption.